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Emotions in Sport Practice

An Integrative Review of Perspectives from Motor Behavior, Sport and Educational Psychology

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Despite long-standing research around competition, the multifaceted role of emotions within sport practice remains underexamined. This narrative review integrates literature from across psycho-motor disciplines in kinesiology to outline how emotions are implicated in various practice goals. We focus on how emotions may enhance the quality of sport practice in support of elite athlete development. We highlight how various emotional experiences may facilitate transfer of learning, then more fully consider how emotions serve the development of athletes' 'learning-to-learn' skills and their motivation to practice. Overall, our aim is to initiate dialogue on the role and complexity of emotions in sport practice. In doing so, we propose that emotions serve conceptually different roles depending on whether the preeminent practice goal is competitive transfer, equipping athletes with learning proficiencies, or maintaining/sustaining athletes' motivational beliefs. We hope that future research and application may consider these perspectives relating to the development of sport expertise.

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Introduction

Emotions can powerfully influence an athlete's attention, cognitions, actions, motivations, and performance (Janelle et al., 2020; Jones, 2003). In psycho-motor research on emotions in sport, most scholarship has considered emotions in relation to competitive performance. Very little research has explicitly considered emotion in relation to sport practice, and specifically how athletes navigate emotions in dedicated learning and skill acquisition activities. However, scholars who have a particular interest in practice conditions as a primary condition for elite athlete development have started drawing attention to this lacuna. For example, Headrick and colleagues (2015) advocated that practitioners consider the role of athletes' emotions in simulating competition experiences during practice, while accounting for how emotional tendencies evolve as an athlete develops. McCardle et al. (2019) called for greater attention to emotional self-regulation especially when athletes are managing aspects of their own deliberate practice activities. Hodges and Lohse (2022) proposed that emotions play a crucial mediating role in how athletes take on more difficult practice, particularly while negotiating threats to their motivation. Wulf and Lewthwaite (2016) contended that practice conditions which promote positive affect crucially enhance athletes' motivation and, ultimately, their motor learning. Though these perspectives draw from different fields of psycho-motor behavior science, efforts to integrate them are rare.

The aim of this narrative review is to provide an overview of the multifaceted role of emotions relating to sport practice, by integrating emerging research from various fields of psycho-motor behavior. This narrative is distilled through a lens on *elite athlete development*, which is also referred to as the study of sport expertise (see Farrow et al., 2017; Starkes & Ericsson, 2003; Williams & Ford, 2008). The study of sport expertise borrows on traditions from sport psychology and motor learning, with a particular emphasis on how skill acquisition occurs. Notably, it also implicates conceptualizations of skill development and athlete learning from a psycho-pedagogical perspective from education psychology (Zimmerman, 2006). In terms of narratives around elite athlete development, practice conditions have long been at the fore (e.g., Ericsson et al., 1993; Hodges & Starkes, 1996). Young, Baker et al. (2025) reviewed broad literature to conceive a model of inter-individual variability in elite athlete development, which emphasized how talent development results from the influence of psychological constructs on athletes' accrual of practice amounts over time, as well as how they serve to optimize "facets of quality sport practice" (p.142). However, other than brief mention of emotional responsiveness as a variable relating to practice, little attention was granted to emotions. Consequently, in this review, we interrogate the role of emotions in enhancing the quality of sport practice.

We propose that to better understand the training conditions that optimize the quality of sport practice, emotions need to be considered alongside predominant practice goals. We begin by reviewing definitions of emotion before examining how emotions have been situated within traditional motor learning theories, advancing the role of emotionally laden practice for transfer of learning. While transfer of learning to competition is often a preeminent practice goal, we argue that emotions serve other objectives that are important in their own right, such as enhancing athletes' acquisition of 'learning-to-learn', self-regulatory proficiencies. We then discuss emotions in service of motivation to practice/train, discussing the value of flow-like conditions in specific practice activities, and locating emotion as both an antecedent and consequence of meaningful outcomes during sport practice. This review is based on literature from different theoretical principles relating to emotions, with various implications in relation to different types of sport practice. Table 1 offers an overview of this information and serves as a map of our progression through this literature in relation to predominant types of goal-oriented practice.

Table 1: Overview of how emotions are situated within predominant types of practice goals, according to theoretical principles, along with theoretical implications.

Predominant Practice Goal	Underlying Theoretical Principle (Home Discipline)	Primary Reference(s)	Theoretical Implications
Using Practice Conditions to Facilitate Skill Transfer	Specificity of Learning Hypothesis (Motor Learning)	Henry (1968); Lawrence et al. (2014)	There should be emotional similarity between practice conditions and competition contexts.
	Affective Learning Design (Ecological Dynamics)	Headrick et al. (2015)	There should be emotional similarity between practice conditions and competition contexts.
	Individualized Zone of Optimal Functioning (Applied Sport Psychology)	Hanin (2000, 2007)	The emotional preferences of an athlete are likely to vary between practice and competition contexts (i.e., dissimilarity).
Using Practice Conditions to Develop Proficiencies that Support Skill Development	Challenge-Point Framework (Motor Learning)	Guadagnoli & Lee (2004); Hodges & Lohse (2022)	Emotions arising from challenging practice conditions should be balanced with the motivational costs of learning in those conditions.
	Self-Regulated Learning (Educational Psychology; Sport Psychology)	Young et al. (2024); Zimmerman (2006)	Emotions are intertwined with key learning-to-learn, self-regulatory processes activated around practice. Emotions can catalyze self-regulatory processes, and emotions are a consequence of self-regulatory processes.
Using Practice Conditions to Promote Motivation & Encouragement	Flow (Sport Psychology)	Jackson & Csikszentmihalyi (1999)	Flow-like practice conditions can promote emotions that reassure/build an athlete's confidence and sustain engagement.
	Attribution Theory (Sport Psychology)	Hanrahan & Biddle (2008); Weiner (1985)	Emotions invite an athlete to appraise the nature of their practice outcomes. The appraisal of outcomes results in facilitating/debilitating emotions, with implications for practice motivation. Attributional retraining may help athletes navigate challenging emotions during practice.
	OPTIMAL Theory (Motor Learning)	Wulf & Lewthwaite (2016)	Practice conditions that enhance expectancies improve performance and learning and are accompanied by positive affective experiences.

Methods for Identifying Key Literature on Emotions and Practice

As no one field appears to have uniquely addressed athletes' emotions during practice, it was important to integrate pertinent literature from various fields, including sport psychology, motor learning, ecological dynamics, and educational psychology. Narrative reviews are valuable for synthesizing information from sources that use diverse methodologies and that draw from different theories and/or bodies of literature (Baumeister & Leary, 1997), without requiring a formal systematic approach for identifying pertinent literature (Grant & Booth, 2009). We felt this approach was appropriate given the breadth of research designs in these kinesiology-related fields, and our aim to merge ranging narratives to the topic of elite athlete development.

Our initial engagement with this topic began by reviewing formative theoretical literature on emotions in sport, such as Hanin (2000), Lazarus (2000a), and Vallerand and Blanchard (2000). We then reviewed contemporary literature from skill acquisition (e.g., Headrick et al., 2015; Hodges & Lohse, 2022) and sport expertise (e.g., Baker et al., 2020; Young et al., 2024) lenses to gain a sense of how emotions have been situated within these domains as it relates to practice. We built our search outwards using secondary iterative searches within these bodies of literature, which implicated readings from educational psychology (e.g., Efklides et al., 2017; Zimmerman, 2006) addressing emotions during the negotiation of learning activities. Our reading revealed the need to review and understand literature that addressed contextual nuances as it relates to the role of emotions across differing practice goals (e.g., Hanin & Syrjä, 1997), leading us to frame parts of the literature review around types of practice (Hodges & Lohse, 2022). Ultimately, convergence of these bodies of literature and their associated texts offer a comprehensive and integrative overview of emotions in sport practice.

What Are Emotions?

Emotion is an affective state experienced in reaction to a specific event/stimulus, that is often subject to cognitive appraisal, and which leads to various physiological, cognitive, and behavioral changes (Janelle et al., 2020; Vallerand & Blanchard, 2000)¹. Research has typically conceptualized emotions from two main perspectives: a *dimensions approach*, emphasizing underlying properties of emotions (e.g., Lang et al., 1997), or a *discrete approach*, arguing that emotions are qualitatively distinct (e.g., Izard, 1991; Lazarus, 2000a).

Regarding the former perspective, emotions are described according to two prominent dimensions: *affective valence* (pleasure/displeasure) and *activation* (low to high arousal) (Lang et al., 1997). Affective valence serves to activate either our appetitive (in response to pleasant stimuli, attracting one to a task) or defensive response system (in response to threatening stimuli, preparing one to flight, flee, or freeze) (Janelle et al., 2020). The activation dimension further modulates processes associated with attention, perception, and action (Janelle et al., 2020). Some theorists have argued that affective and activation valence may not fully encompass the action tendencies of an emotion and have advanced a third dimension, *motivational direction*, which captures avoidance/approach action tendencies (e.g., Carver & Harmon-Jones, 2009).

The discrete approach emphasizes that distinct emotions have unique accompanying experiences resulting from one's appraisal of a situation (Vallerand & Blanchard, 2000). For instance, Lazarus (2000b) contended that cognitive-motivational-relational experiences arise as part of an ongoing, two-part appraisal process of potential threats/demands in the environment. This appraisal process leads to a distinct emotion which has its own unique pattern of cognitive, physiological, motivational, and behavioral consequences. During primary appraisal, an individual discerns whether anything is at stake and whether a situation is harmful, beneficial, or challenging - giving rise to a specific, discrete emotion (e.g., fear, hope). This distinct emotion in turn has a central meaning, or core relational theme, that encompasses the appraisal process and has implications for an accompanying action tendency. For instance, the core relational theme derived from the appraisal process for anger is "a demeaning offense against me and mine" while anxiety is associated with the core relational theme of "facing uncertain,

¹We recognize broader discussion over the definition of emotion and refer readers to Izard (2010) for fuller treatment. In this paper, we use the term affect as a more generic term that encompasses emotions, feelings, and moods in contrast to emotion which is state specific. Mood refers to lower intensity affective states that tend to last for longer durations and typically do not have a discernable stimulus. Feelings refer to an individual's subjective experience of emotions and mood, independent of any accompanying physiological or behavioral changes.

existential threat” (Lazarus, 2000b, p. 234). Their accompanying action tendencies also differ as anger is associated with “counterattacking”, whereas the consequence of anxiety is to avoid or escape. Such distinct emotional responses are then regulated through a person’s secondary appraisal, whereby they judge their options/resources for coping relative to the perceived threat/demand, which may temper and control the associated action tendency.

How Do Emotions Experienced in Practice Better Prepare Athletes for Competition?

Discourse around sport practice commonly centers on the expectation that improvements made during training should effectively transfer to competition. In the motor learning domain, significant distinction is made between measuring performance during skill acquisition (i.e., performance of a motor task during practice blocks) and assessing learning (i.e., improvement in performance on a task as result of practice). Specifically, learning is assessed either through retention tests (i.e., performance on the same task following a period of no practice that is typically ≥ 24 hrs) and/or transfer tests (i.e., performance on some sort of variation of the practice task also following a period of no practice) (Schmidt et al., 2019; Schmidt & Bjork, 1992). Thus, the value of any practice condition/variable is predominantly judged by how it influences a) retention or the relatively permanent improvement in performance, and/or b) the robustness of performance to transfer to other conditions/variations.

The extent of transfer of learning depends on the *degree of similarity* between practice tasks and criterion tasks in the other conditions/variations (Schmidt et al., 2019). Different motor learning theories emphasize distinct dimensions of similarity, such as identical elements theory (Thorndike, 1914), transfer appropriate processing theory (e.g., Lee, 1988; Morris et al., 1977), though they typically neglect considerations of affective components. However, only the *specificity of learning hypothesis* (Henry, 1968), which suggests that effective transfer depends on similarity in sensorimotor demands, extends to note the role of anxiety in facilitating transfer from practice to criterion conditions. For instance, using the specificity of learning hypothesis, Lawrence et al. (2014) found that mismatches in anxiety between practice and criterion tasks attenuated transfer. Specifically, they found that training exclusively under either high or low anxiety improved performance only in matched criterion conditions, whereas training with mixed high and low anxiety enhanced transfer to both high and low anxiety criterion conditions.

With respect to emotional similarity and transfer of learning, much of the existing literature has centered around anxiety.² A recent meta-analysis found that embedding high anxiety into practice conditions (termed pressure training) consistently improved performance on criterion conditions of high anxiety (akin to competition conditions) when compared to groups that trained under normal/low anxiety levels (Low et al., 2021). While this pressure training literature seems to support the similarity principle, there are a few caveats. First, most studies used high anxiety transfer tests, rather than real world competitive scenarios, so questions remain about how embedding anxiety in practice will transfer to competition (Low et al., 2021). Second, while the intensity of certain competitive affective states might be difficult to mimic in training (e.g., the pressure of an Olympics event), there is evidence that training with moderate anxiety levels can enhance transfer to higher anxiety criterion contexts (Cabral et al., 2024; Oudejans & Pijpers, 2010). Therefore, while the presence of certain discrete emotions during practice, such as anxiety, might be important for transfer in conditions implicating the same emotion, the identical intensity of that emotion might not be as essential to facilitate transfer. Third, Lawrence et al. (2014) found that pressure training was most effective when anxiety was introduced later in the practice sequence, suggesting that its use should be carefully timed and possibly integrated alongside other learning-related goals. However, although anxiety introduced at a later stage in practice served to facilitate transfer to criterion tasks under anxiety, the introduction of anxiety during practice also impaired practice performance. Thus, pressure training may potentially disrupt key skill acquisition processes, may interrupt automaticity,

²This focus on anxiety as the embedded emotion is likely due to a long tradition of prioritizing anxiety in studies of sport emotions due to its disruptive effect on performance (e.g., Cottyn et al., 2012; Pijpers et al., 2005, 2006; Runswick et al., 2018) and pre-eminence in discussions of choking in competition (see Janelle et al., 2020; Oudejans & Nieuwenhuys, 2009).

and/or disrupt confidence during practice blocks. Therefore, this phenomenon raises questions about the trade-offs involved in embedding anxiety into skill acquisition trials or practice blocks.³

Ecological dynamics, which defines similarity in terms of perception and action coupling (Davids et al., 2013; Pinder et al., 2011), has extended the principle of emotional similarity to include a broader range of emotional experiences (Headrick et al., 2015). Recognizing that athletes experience a gamut of emotions beyond anxiety (e.g., Martinet et al., 2012), the concept of affective learning design examines how affective experiences, including emotions like dejection, happiness, anger, excitement, and anxiety, influence transfer (Maloney et al., 2018). These emotional experiences are considered from an individual perspective, acknowledging that emotional tendencies differ between athletes and evolve over time. Consequently, it is important to create practice environments that allow for the experience/expression of individual emotional tendencies that are representative of competition (Headrick et al., 2015).

In summary, leaning on literature from motor learning and ecological dynamics, the role of emotions is foremost considered with respect to how practice conditions facilitate transfer to competition. When the transfer of skills from practice to competition is the foremost criterion that practitioners use to shape emotionally laden practices, the literature supports that emotional experiences should correspond between practice and competition. However, we question whether transfer is always the foremost criterion by which we should evaluate the qualities and effectiveness of training, and wonder whether emotional similarity alone encourages a limited view to how athletes navigate the experience of practice. With this in mind, we now turn to other literature that addresses contextual nuances and athletes' preferences around emotions, integrating a broader perspective.

How Might Athletes Benefit from Varied Emotional Experiences During Practice?

In applied sport psychology, the Individual Zones of Optimal Functioning (IZOF) model (Hanin, 2000, 2007) is a prominent and useful framework for understanding emotions and athlete performance. It posits that each athlete has an emotional profile at which they perform best. This IZOF profile describes an athlete's ideal affective profile through five dimensions: form, content, intensity, time, and context. Form represents the subjective experience of affect, including cognitive, motivational, biological, and behavioral components. Content describes the qualitative characteristics of specific affective states in terms of hedonic tone (pleasure-displeasure) and functional impact (whether that emotion/feeling optimizes performance or is dysfunctional and hinders it). Intensity captures whether an athlete's experience of an emotion/feeling is high, low, or moderate, and whether the level is 'in' or 'out' of one's optimal zone for performing. Time reflects the dynamic changes of affective states relative to the timing of one's performance.

Notably, Hanin's (2000) model predicts that an athlete's personalized IZOF profile, and specifically their preferences for emotions, should change as a function of environmental *context*. Thus, the emotions that an athlete deems as being functional/dysfunctional will change between practice and competition settings, as will their preferences for a desired intensity of a given emotion. Although this is a notable proposition, only one study has compared athletes' emotional preferences between practice and competition. Hanin and Syrjä (1997) compared the emotional profile of 12 Olympic-level Finnish Nordic skiers across three settings: competition, hard training, and technical skills training. At different points in a ski season, each skier was asked to create a list of positive (e.g., active, calm, confident, determined) and negative (e.g., nervous, angry, annoyed, dissatisfied) emotions and feelings that were facilitating and debilitating for each of the three settings, and to identify the intensities at which they felt those affective states were helpful or harmful. Analyses focused on determining the degree of similarity between the different settings. Results indicated a moderate overlap in athletes' list of preferred emotions between competition and hard training ($M = 41\%$), with lower overlap between technical skills training and competition ($M = 27\%$), and between technical skills training and hard training ($M = 31\%$). Further, the athletes reported higher preferences for the intensity of positive, facilitating emotions in competition than in both types of practice. Additionally, negative emotions were viewed as harmful if they were at lower intensities in competition, than in both practice contexts. These results confirmed IZOF model predictions, that is,

³We recognize that well-designed retention and transfer tests are crucial for distinguishing learning from temporary practice effects (Schmidt & Bjork, 1992). Here, however, we question how the timing of practice interventions like pressure training may influence athletes' psychological processes for competition readiness.

athlete preferences for emotions not only differs between practice and competition, but between different types of practice.

The contextual proposition of the IZOF model questions the principle of emotional similarity that dominates within motor learning and ecological dynamics. Hanin and Syrjä's (1997) findings, and applied practice with the IZOF in years since (which has upheld the contextual proposition, see Hanin & Ekkekakis, 2014) proposes differences in athletes' preferences according to context. Thus, there are questions: whether emotional similarity (underpinning transfer) is always necessary, and whether there are instances where applying emotional similarity might hinder the benefits athletes might gain from practice?

Why Might Athletes Have Varying Preferences for Emotions in Practice?

The findings from Hanin and Syrjä (1997) suggest that the type of practice may moderate emotional preferences. In this vein, Hodges and Lohse (2022) recently began to unpack the value of different types of practice, tendering the notion that different types of practice can have different goals. They provided a useful heuristic in differentiating practice categories based on their specificity relative to competition and the relative degree of challenge during a practice session. In addition to "practice to transfer", they introduced categories entitled "*practice to learn*" and "*practice to maintain*" (p. 762). In these latter two categories, different practices have inherent goals of their own, *rather than solely serving the criterion of transfer*. Pertinent to our review, the implication is that transfer to competition is not the only criterion for understanding emotions during practice. Instead, emotions are seen as an inherently important part in facilitating athlete development irrespective of transfer, which means one can consider alternative goals of practice and emotions in new ways.

How Are Emotions Involved in Practice to Learn?

According to *practice to learn*, a practice can be inherently valuable if it provides athletes with new information that they can use to guide further skill development (Hodges & Lohse, 2022). As designers of the training environment, coaches may at times forgo elements of specificity and sacrifice in-practice performance to provide athletes with novel and challenging information that can help in their activities of skill development. Drawing from the Challenge Point Framework (Guadagnoli & Lee, 2004), practice to learn focuses on how coaches can modify the relative challenge of a practice by manipulating different practice variables (e.g., variability and feedback scheduling) to create an optimal challenge zone that is slightly beyond an athlete's current skill level. In this type of practice, coaches want athletes to make mistakes, requiring them to engage various cognitive and motivational strategies involved in planning, analyzing, and evaluating performance (Hodges & Lohse, 2022). Engaging and *learning to engage* these cognitive and motivational proficiencies are believed to lead to more sustained learning (Wadden et al., 2019; Young et al., 2024).

Due to the decrement in overt performance indicators during practice to learn, this type of practice might not be inherently motivating. As such, within practice to learn activities, emotions are largely considered as negative affect associated with the motivational consequences of making mistakes (Hodges & Lohse, 2022). In this narrative, coaches orchestrating practice to learn settings need to consider conditions that support further development of psycho-motor skills and proficiencies, which involves exposure to errors balanced against the challenging emotions those errors can evoke. Literature suggests that there may be individual differences in affinity towards differing practice difficulties, with many learners being drawn to practices of easier difficulty as a way of supporting competence (Kool et al., 2010; Simon & Bjork, 2001). However, notably, there is evidence that elite athletes elect to structure their practice environments in ways that promote greater challenge, cognitive effort, and emotional demands than less elite athletes (Coughlan et al., 2013), which may suggest an affinity for elite athletes in pursuing practice to learn activities.

The idea that emotions might be associated with athletes' 'learning-to-learn' proficiencies is also found in recent work in sport psychology known as self-regulated sport practice (e.g., Young et al., 2023). Within this literature, athletes' psychological processes are examined to understand their role in enhancing practice efforts and cognitive engagement in the process of sport practice. There is evidence for key

athlete-initiated processes, called self-regulated learning (SRL) processes, that are critical for raising athletes' self-awareness of their control over their cognitions, actions, motivations, and *emotions* while they pursue practice goals (McCardle et al., 2019; Zimmerman, 2006).

Despite their noted role in athletes' self-regulation during training, emotions have been relatively overlooked within the self-regulated sport practice literature. Recent scholarly work has begun to address this oversight. For instance, Wilson et al. (2021) suggested that feelings of challenge could prompt athletes to engage various SRL processes, such as planning, goal adjustment, and/or reinvestment of effort on a task. This led Young et al. (2024) to contend that certain affective states (e.g., feelings of challenge and frustration) could catalyze the enactment of SRL processes during arduous training. In particular, the rise of poignant affective states was proposed as being instrumental in 'turning on' SRL processes, or alternatively 'dampening' SRL processes during instances in training (Young & Siekańska, 2024). Additionally, interviews with sport psychology practitioners have identified emotional regulation as an important target of SRL by adolescent athletes in relation to quality sport practice (Young, Bain, et al., 2025). Specifically, emotions were described as intertwined with a) planning strategies prior to practice trials; b) strategies for controlling, self-monitoring, and adjusting during practice trials; and with (c) post-task reactions/reflections that guide athletes towards future trials/sessions.

These insights regarding emotional regulation and 'learning-to-learn' proficiencies are more prominently featured in parallel SRL work from educational psychology. For instance, within the Metacognitive and Affective Model of SRL (Efklides et al., 2017), affective feelings, such as enjoyment or anxiety, are discussed as influencing an academic learner's motivational direction and decision to initiate a practice activity. During those practice activities, feelings and emotions, such as surprise or frustration, may disrupt a learner's current processing, directing their attention towards aspects of a task that they need to adjust. Beyond this model, other educational psychology scholars emphasize the role of emotions in influencing learners' engagement around their practice activities. Zimmerman (2006), for example, discussed how learners can use SRL strategies to induce positive affective states prior to learning, which facilitates better practice performance, while Panadero (2017) described how learners' self-evaluation of performance during practice activities give rise to feelings of satisfaction or disappointment that have further motivational implications.

Among scholars who adopt a social-cognitive lens on SRL, there has been little emphasis on these planful and self-reactive roles of emotion as they relate to sport practice. Research that documents how athletes prefer to feel during tough training, and how they negotiate difficult emotions around mistakes or during learning plateaus, are also an under-examined element of SRL. Yet, such emotions, and affective preferences, are likely poignant in athletes' efforts to train during Hodges and Lohse's (2022) *practice to learn* activities. Further, although emotions are presumably involved in enacting key SRL processes during practice, to our knowledge, work has yet to consider the cognitive-motivational-relational consequences of distinct emotions in practice. For example, how does hope, disappointment, guilt, or embarrassment facilitate or debilitate athletes' self-control and action around intensive sport practice? Our position is that emotional self-regulation is an important, multifaceted proficiency for engaging in quality sport practice. Practice in competitive sport inevitably involves salient emotions, thus, learning how to negotiate and exploit these emotions is critical to shaping high quality practice efforts over time. Therefore, irrespective of whether a distinct emotion is experienced in competition, athletes will need to acquire proficiencies for tackling and benefiting from tough/intense practice; the benefits of managing such conditions are critical for skill acquisition (Ericsson et al., 1993).

How Are Emotions Involved in Practice to Maintain?

The objective within Hodges and Lohse's (2022) *practice to maintain* is to build athletes' confidence and allow for automaticity of motor skills. To encourage this type of practice, coaches are urged to match the challenge of a given practice within the bandwidth of an athlete's skills and expectations for success, leading to a high amount of success in the practice. Whereas in *practice to learn*, difficulty was slightly beyond an athlete's current skill level, here the skill-challenge balance is closely matched to allow athletes to demonstrate their skills, reinforce their strengths, develop confidence, and maintain their current performance level through consistent execution (Hodges & Lohse, 2022).

In some respects, this type of practice aligns with longstanding characterizations of flow - an optimal psychological state that is marked by feelings of mental and physical preparedness, confidence, feeling

relaxed and energized, and having total focus on an activity (Csikszentmihalyi, 1990; Jackson, 2007). Specifically, flow is predicated upon an athlete's perceived 'balance' between skill level and challenge; if the challenge of an activity is too easy, an athlete might feel bored and apathetic, and if it is perceived as too hard, the athlete will feel overwhelmed and anxious (Jackson & Csikszentmihalyi, 1999). Moreover, like practice to maintain contexts, flow-producing conditions are characterized by reduced deliberate self-regulation or diminished cognitive effort related to SRL processes. Making mistakes is antithetical to the flow experience because they lead to self-evaluations, causing one to become self-conscious, interrupting the feelings of effortlessness and automaticity that encapsulate flow (Jackson, 2007). During flow, athletes feel that they are demonstrating and exercising their competence without the need for deliberate thought, reflection, or conscious control (Jackson & Csikszentmihalyi, 1999); contrasting it to practice to learn, which we contend necessarily implicates SRL processes.

We propose that emotions of joy and happiness fit well with practice to maintain due to their correspondence with flow. Joy/happiness is a positive emotion that is characterized by making progress towards one's goals (Csikszentmihalyi, 1990; Lazarus, 2000b). Flow experiences require individuals to be so immersed in the enjoyment of the task that nothing else seems to matter. This is not to suggest that practice tasks producing enjoyment and flow are easy; rather, the effort involved in these tasks aligns with a balance between skill and challenge. Whereas in our conceptualization of practice to learn contexts, pushing beyond the skill-challenge balance was seen to evoke challenge-related emotions that prompt the activation of learning-to-learn skills, appropriately challenging tasks in practice to maintain settings help to foster flow and enjoyment, while building confidence and self-assurance.

While we have suggested enjoyment as a prototype of flow fits with practice to maintain, flow researchers have exclusively discussed flow with respect to competition and not within the practice setting. This is a conspicuous omission, considering Csikszentmihalyi's (1990) original contention that flow is implicated across a range of contexts, including practice. Further, educational research shows that there are important implications that can arise from learning conditions that promote flow, even if flow is not necessarily experienced. For instance, when learning conditions involve students practicing within the proper bandwidth of skill-challenge balance, they demonstrate more on-task practice behavior (Tousignant & Siedentop, 1983), more enjoyment, greater learning engagement, and higher levels of perceived control (Shernoff et al., 2003). Thus, whether flow is experienced or not, flow-producing conditions, which are largely viewed as controllable by athletes (Swann et al., 2012), may be inherently valuable conditions for practice and learning (Custodero, 2002). Specifically, emotions evoked by flow-producing practice conditions may help athletes to maintain an elevated locus of control over how they exercise their sport competencies, and motivation.

On this point of motivation, feelings of fluidity, automaticity, and competence experienced during practice to maintain activities are presumably important for fostering short-term readiness, self-assuredness, and confidence ahead of upcoming competitions (Hodges & Lohse, 2022). Moreover, the positive affective experience of practice to maintain is also likely essential for sustaining long-term practice engagement. For instance, within academic settings, students who are outside of their proper skill-challenge bandwidth (i.e., are consistently over or under-challenged), experience boredom, and detriments to long-term motivation and career aspirations (Acee et al., 2010; Krannich et al., 2019; Liu et al., 2013). Overall, while negatively valenced affective states such as confusion, frustration, and anxiety (which we equate to practice to learn conditions) might have some utility in promoting student learning in certain contexts/episodes, positively valenced emotions (which we align with practice to maintain) are essential for facilitating learning over longer periods of time (Tan et al., 2021). The most effective 'mix' of affective states drawn from practice to learn and practice to maintain has not been examined in sport training; yet understanding how athletes balance the emotional substrates of 'learning-to-learn' proficiencies with those that sustain self-confidence and self-assuredness is an important area of future research.

How Are Emotions and Motivation Intertwined in The Pursuit of Quality Sport Practice?

Emotion and motivation are closely intertwined, with prominent emotion theorists situating emotion as both antecedents and consequences of emotion (Vallerand & Blanchard, 2000). For instance, Izard (1991) argued that emotions are the primary system by which a person organizes their motivations, while

Lazarus (2000a) emphasized that emotions arise in reaction to the status of one's goals. While emotions are likely important for directing an athlete's efforts towards a sporting pursuit (at a domain-level of motivation), this section will primarily focus on the involvement of emotions and motivations as they relate to practice scenarios or training tasks (see Young, Baker, et al., 2025). We consider how emotions are implicated in motivated efforts to optimize the quality of one's practice efforts, and in sustaining an athlete's engagement such that they can accumulate greater practice volumes.

One prominent theory that could help better understand how emotions are involved in the regulation of motivation and effort during sport practice is *attribution theory* (Weiner, 1985). In short, attribution theory focuses on how athletes make sense of personal outcomes/events in sport, and how this sense-making relates to subsequent motivation. Attribution sport research almost exclusively focuses on competition outcomes (Biddle et al., 2001; cf. Cleary & Zimmerman, 2001), though it could be used to richly understand outcomes following practice trials or sessions. Within attribution theory, emotions feature prominently (Hanrahan & Biddle, 2008; Weiner, 1985). For instance, an athlete may have expectancy-related emotions (i.e., hope, anxiety) heading into challenging practice activities that are shaped by their goals, perceived level of control, and prior success. Following outcomes of those practice activities, and prior to any attributions, the athlete will experience basic emotions such as happiness, frustration, or sadness. If practice outcomes are unexpected, heavily valenced, and/or personally important, the athlete will initiate a causal search for why those outcomes occurred. If an outcome is attributed to internal causations (e.g., personal ability), self-conscious emotions, such as shame or pride may be elicited. Feelings of helplessness or hopefulness may be evoked depending on whether an outcome is viewed as being fixed or changeable. Finally, depending on the perceived controllability of that event, an athlete may feel pity (e.g., when they lack resources to respond to a negative outcome), gratitude (e.g., when they attribute a positive outcome to their actions/decisions), or guilt (e.g., when they judge a negative outcomes as a consequence of something they could have done, but failed to do). These emotional responses may be accentuated among athletes who aspire to higher talent levels, because there is much at stake (personal meaning) during their practice activities.

Athletes' attributions for success and failure shape their emotions and should in turn have motivational consequences impacting their capacity beliefs and striving behavior (Hanrahan & Biddle, 2008) in practice. Thus, attributional retraining likely represents an important avenue for the development of quality sport practice habits. For instance, training learners to make attributions following mistakes that are more internal, unstable, and controllable should improve performance in subsequent practice trials, foster more positive and facilitating emotions, also resulting in higher expectations, self-efficacy, greater persistence, and re-investment of effort on future trials (Biddle et al., 2001; Cleary et al., 2006; Cleary & Zimmerman, 2001; Fösterling, 1985). Importantly, the emergence of greater effort and self-efficacy following adaptive attribution-making is a significant facilitator of greater practice quantities and enhanced practice quality (Zimmerman, 2006). Therefore, it stands to reason that retraining attributional processes to elicit specific emotions, particularly those that facilitate greater investment or strategic behaviors, or do so in a reparative way (e.g., associated with guilt about an outcome, Izard, 1991), may be important for enhancing motivational processes around practice.

This narrative of how emotions and motivational processes are implicated within practice and learning has also begun to receive attention in the motor learning literature, as exemplified by Wulf and Lewthwaite's (2016) *OPTIMAL theory* (Optimizing Performance through Intrinsic Motivation and Attention for Learning). In short, this theory posits that practice conditions that enhance an athlete's motivational beliefs, particularly their expectancies for future performance, will help to facilitate greater performance and learning, and create positive affective experiences. However, this theory does not explicate the relationship of affect to performance or learning. Although the motivational claims of OPTIMAL theory (i.e., whether the specific practice conditions consistently lead to improved motivation, and whether motivation mediates learning) are contested (for reviews see McKay et al., 2023, 2024; Parma et al., 2024), alternative explanations for how these practice conditions enhance learning nonetheless implicate affective phenomena. For instance, reinforcement learning frameworks attribute learning to reward-based mechanisms (Lohse et al., 2020) implicate affective responses such as self-satisfaction (Sander & Nummenmaa, 2021). Even in models that do not foreground motivation, affective responses remain critical in shaping how athletes engage with and benefit from practice; yet these affective processes are still not adequately represented in the sport practice literature.

Thus far, we have discussed how emotions have an important role in catalyzing various motivational processes and beliefs that support skill development during practice. While some theories emphasize the value of positive emotions that accompany practice conditions (i.e., flow, OPTIMAL theory), others recognize the importance of more difficult, challenge-related affective experiences (i.e., scholarly works implicating SRL processes for enhancing practice). This tension between negative/challenging and positive emotions harkens back to the deliberate practice theory of talent development (Ericsson et al., 1993), which contended that the most beneficial practice conditions for skill acquisition should be challenging and not inherently enjoyable. Subsequently, there were notable debates within sport about whether deliberate practice was always emotively challenging (see Baker et al., 2020), or whether athletes could conceive of deliberate practice in more enjoyable or favorable emotional states (e.g., Hodges et al., 2004; Hodges & Starkes, 1996; Young & Salmela, 2002). Based on this narrative review, we believe the contemporary tension that may need to be resolved is between how much practice should implicate negatively valenced affect (which can be commensurate with striving and contending to meet a challenge) and how much practice should implicate positively valenced affect (commensurate with exercising self-competency or feeling flow-like states as one successfully meets the challenge). In the resolution of such tension, it is important to recognize that emotions plausibly intersect with the types of goal-oriented practice introduced by Hodges and Lohse (2022). The type of practice will determine the extent to which positive affect is the goal or the target of intentional practice design, and the extent to which intensive metacognitive engagement need accompany practice design.⁴ Resolving the positive versus negative/challenging tension as a function of practice type may be important for understanding the interplay of emotions and motivational processes that underpin efforts oriented toward high-quality practice, and for sustaining the motivation required to accumulate greater volumes of training necessary for long-term development.

Summary and Future Directions

In this review, we integrated theory and research from literature in motor learning, sport and educational psychology, to provide various perspectives on the multifaceted role of emotions in sport practice. Without denying the likely importance of emotional attributes during practice for the sake of transfer, altogether, the literature suggests that emotions may have nuanced roles in facilitating alternative goals relating to the process of practice *per se*. Moreover, the literature suggests that the influence of emotions should further differ between types of practice. Similar to how emotions are considered with respect to optimal functioning in competition, there is a need to recognize that athletes have individual preferences (Hanin, 2000) and tendencies (Headrick et al., 2015) that could be better considered for optimal practice functioning.

While our goal with this review was to initiate inter-disciplinary dialogue about the role and complexity of emotions in sport practice, we suggest that a historical barrier to this dialogue is the variety of methodologies used across these disciplines. For example, within educational psychology and sport psychology, there is a reliance on self-report survey-based or interview studies from students and athletes that tend to be descriptive, correlational or quasi-experimental at best (often featuring less internal control). In applied sport psychology, where there is great focus on external validity, practitioners' accounts, narratives, and reflections provide rich insights into the emotional complexities of actual sport practice, yet with limited causal inference. In contrast, motor learning research heavily relies on classic retention/transfer designs that inform highly controlled intervention studies. However, this approach tends to neglect the various emotional demands that may emerge during actual sport practice. Therefore, to effectively move our narrative review from insular, multiple disciplines to true inter-disciplinary approaches, theoretical principles and methodological approaches will need to be reconciled in future inquiry.

Within future research, we encourage researchers to more thoroughly inventory the full gamut of affective experiences that occur during sport practice. For instance, motor learning research has had a biased

⁴We acknowledge within the broader sport literature that other goals may be affixed to practice that sustains athletes' motivation over time. For example, the social environment may be intentionally considered to elicit more positive affect (Baker et al., 2020), including positive coach-athlete interactions or positive synergies between training/teammates. The structuring of the practice environment to promote social elements for enjoyment might be important for keeping athletes engaged in training over time (Chambliss, 1989). We have not elected to unpack this further, as introducing a social element adds a new layer of interpretation to the literature that is beyond the scope of the current review.

focus on anxiety as a dominant affective state, while overlooking other complex emotions that may be influential. Similarly, research on athletes' SRL processes during sport practice has treated emotions implicitly or has referred to emotional regulation in broad terms; as such, there is need for greater precision regarding specific emotions. As athletes' skill acquisition efforts are in flux - in the lead-up to practice, during practice trials, and on exit from practice - future longitudinal sampling would provide valuable insights into how emotions and their effects change. Further, it would be important to examine how emotions change temporally in keeping with changing goal objectives around practice. For example, coaches may assign athletes segments for transfer, segments for learning-how-to-learn through errors, and for reassuring/building readiness, all within the same practice; there is a need to examine how emotional responsiveness/patterning would similarly change. Future research should examine specific momentary emotional experiences, through immediate reflections or real-time ratings during practice, to investigate which emotions emerge during practice and how they relate to athletes' goal-directed cognitions, actions, and motivations.

Based on this review, we predict that different emotions may be facilitative/debilitative for practice depending on the specific goal of a practice (i.e., facilitating transfer, developing and activating learning to learn proficiencies, or motivating and encouraging). We further predict that there should be substantial inter-individual variability in emotional patterns around each type of practice that could be submitted to criterion tests associated with practice behaviors (e.g., effort, persistence, self-correcting, disengagement), approach versus avoidance indices of motivation, and cognitions (e.g., distracted thoughts, attentional focus). We are curious whether these functions occur through the cognitive-motivational-relational theme of individual, discrete emotions (e.g., happiness, pride), or through profiles of co-activated emotions (e.g., an optimal mix, or balanced positive-negative valence). In accordance with conventional sport expertise approaches (e.g., [Baker et al., 2020](#)), we predict that there should be reliable skill level (i.e., novice-intermediate-elite) differences in how emotions influence these criterion variables, and for emotional responsiveness related to outcome attributions in practice. Finally, although we have not addressed it directly in this review, emotions tied to the social presentation of the self and social-evaluative pressures (e.g., in front of coaches, teammates, parents, scouts) may drive distinct emotions or poignantly magnify the influence of emotions around sport practice.

Additional Information

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Approved the submitted version for publication: RR, BWY, SR, MJC

Conflicts of Interest

MJC is currently a section editor for Communications in Kinesiology and previously served as an executive committee member for the Society of Transparency, Openness, and Replication in Kinesiology (STORK). MJC was not involved in any aspect of handling this manuscript except as an author.

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